AI Powered Visa Certification

Advance Machine Learning Easy Visa - PGP-AIML-BA-UTA-Mar25-E

July-12-2025

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Contents / Agenda

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Executive Summary

- The U.S. Office of Foreign Labor Certification processes over 1.6 million visa applications annually, with a 9% year-on-year growth This project aims to develop a machine learning model to reduce OFLC's workload by fast-tracking application reviews and prioritizing **recall**, ensuring maximum identification of qualified candidates while minimizing the risk of overlooking top talent and contributing to skill shortages.

- Applicants with higher prevailing wages tend to have a greater chance of certification, likely due to roles requiring more specialized or scarce skills.

- Large and well-established employers generally enjoy more credibility, whereas startups and smaller organizations face lower approval rates—highlighting a need for deeper analysis to address potential biases.

- Geographic origin of applicants significantly influences outcomes; for example, denial rates are about 42% for South America, compared to only 20% for Europe.

- The West region also shows unique employment dynamics that warrant further exploration.

- Full-time, yearly wage positions dominate certified cases. Standardizing wage units could simplify comparisons and reduce potential biases toward yearly salaries.

- Candidates with prior work experience have a slight edge in certification likelihood.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Problem Overview and Solution Approach

- Office of Foreign Labor Certification (OFLC) tasked Easy Visa to reduce the processing overhead of high volume (1.6 million) positions and workload is increasing by 9% year-on-year basis
- OFLC goal is to make sure employer demonstrate that no sufficient US workers are available to perform the work at the wages that meet or exceed the wage paid for the occupation in the area of intended employment. At the same time OFLC needs to make sure it don't miss on the right talent which can harm the competitiveness or create scarcity of intended skill level
- EasyVisa need to consider OFLC objective of certifying right candidates and comply with Immigration and Nationality Act (INA)
- Machine learning model will help to process high volume of application every year by reducing human intervention, it will also help to remove inconsistency of decision-making and give ability to focus on the right candidates
- Considering these objectives, certifying right candidates and denying unqualified candidates is very important.
- As this is a classification problem Accuracy, Precision, Recall and F1 score will be evaluation criteria
- Precision focus to reduce False Positive or Only recommend qualified Ones
- Recall focus on to reduce False Negative or Try to find all the qualified candidates at the expense of tagging some under qualified candidates being certified
- F1 score tries to balance both the score.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Problem Overview and Solution Approach

- Given the high volume of visa applications and the increasing demand for skilled labor, the selection process must be carefully designed to avoid overlooking qualified candidates—especially those near the decision threshold. In this context, it's more critical to **identify all potentially eligible candidates**, even at the risk of including some borderline cases. Any incorrect approvals can still be filtered through subsequent **human verification**.

  In this solution, the trade-off has been deliberately made in favor of **maximizing recall**. The primary evaluation metric for the machine learning models is recall, ensuring the system captures as many qualified candidates as possible without unnecessarily tightening the selection criteria.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Problem Overview and Solution Approach

- **To build the ML Model following solution steps will be followed**
  - Data Overview and Analysis
    - Load customer data and verify its successful loading through a sanity check
    - check the record size
    - number of columns and data type
    - check some sample data rows
    - Perform univariate and multivariate data analysis
    - Identify and fix data anomalies found in the data
  - Model Evaluation
    - Decide on the type of classification model to consider and perform the below step for each model
    - Select the model scorer (***recall_score***,*f1_score, accuracy_score, precision_score*)
    - Calculate Accuracy, Recall, precision and F1 score
    - Build Confusion metrics
    - Use StratifiedKFold to get mean of cross validated score for each models
    - Build boxplot with the cv score to compare all the models together
    - Evaluate model on train, test and validation datasets and choose the best model

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Business Problem Overview and Solution Approach

- **Model Building**
  - Dataset is clearly labeled, and objective is to classify candidate as either Certified or Denied, To achieve this classification models were explored starting with basic classifier like DecisionTree
  - Various ensemble models explored to reduce bias and variance, as this works better than most single models.
    - Ensemble Model use either averaging various week model (RandomForest, Bagging)
    - Sequential boosting (AdaBoost, XGBoost, Gradient Boost)

- Trained baseline model with original dataset and analyzed their performance
- Applied Oversampling (SMOTE) and retrained models and evaluated results
- Applied Undersampling (RandomUnderSampler) and retrained models and evaluated results
- Used **RandomizedSearchCV** to tune the top -performing models from both sampling approach
- Compared performance across all model-sampling combinations using evaluation metrics (e.g., recall, precision, F1 score).
- Selected the best-performing model based on evaluation metrics (Recall)and generalization

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results

Statistical Insight

- The dataset contains 25,480 records. Most applicants are from Asia (66%) and hold a Bachelor's degree.
- Around 58% have prior job experience, and 89% do not require job training.
  The average number of employees per organization is ~5,667, with some negative values needing correction.
- The average year of establishment is 1979, and prevailing wages vary widely with a mean of $74.5K, ranging up to $319K.
- Yearly wage units dominate the data, and most jobs are full-time positions. About 67% of applications are certified
- Employer Distribution: 75% of organizations have fewer than **3,504 employees**, indicating a skew toward small to medium-sized employers.
- The vast majority of jobs are **full-time (89%)** and listed with **yearly wages (90%)**, which suggests these are standard professional roles.
- About **two-thirds of applicants** are certified, but approval likelihood appears to vary significantly by region, wage, and employer characteristics.

Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results

- **Correlation metrics** reveals negligible correlation among prevailing wage, year of establishment and number of employees as correlation coefficient is near zero. These Features are statistical independent and each provide unique information and can't be used as substitute for others in the model building

- **Education of employee and case status**: Approximately two-thirds of applicants with education up to high school had their visas denied. In contrast, candidates with a Doctorate represent a small portion of the applicant pool and face only a 12% rejection rate. Master's degree holders experience relatively low rejection, while Bachelor's degree applicants face a moderate denial rate compared to higher education levels.

- **Continent of employee and case status**: Region plays a significant role in visa approval outcomes. Applicant from Europe have the highest approval rate, around 80%, followed by those from Africa and Asia. In contrast, applicants from South America face the lowest certification rates. Additionally, candidates from Oceania and Africa are relatively rare in the applicant pool.

- **Prior Job experience:** Applicants with prior job experience not only form the majority of the applicant pool but also have a significantly higher approval rate of around 75%, compared to just 56% for those without prior experience

Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results

Great Learning
POWER AHEAD

- **Prevailing wages vs region of employment**: Across all regions of employment, prevailing wages show significant outliers that impact the mean. The Island and Midwest regions have the highest average wages (~91K), while the Northwest has the lowest (~67K). Interestingly, the South reports higher average wages than the West. Overall, wages range widely—from below 100 to over 300K

- **Prevailing wages and approval status:** Certified candidates show a strong concentration of wages between 60K and 120K, with a right-skewed distribution and a mean of 77K (excluding outliers). Lower-wage applicants are rarely certified, and a noticeable spike at $0 may indicate internships or data anomalies.
- Denied candidates tend to have lower wages, with a flatter, slightly right-skewed distribution and a mean of 68K. A significant spike near $0 is also observed in denied cases, and wages above 200K are rare among them.

  Certified applicants tend to have higher wages, a trend clearly reflected in the box plot—with or without outliers. The mean wage for certified candidates is approximately 77K, compared to 68K for denied applicants. When outliers are removed, the mean drops slightly to around 73K for certified and 66K for denied applicants, reinforcing the positive correlation between higher wages and visa approval.

- **Unit of wages and case status**: Approximately 90% of jobs offer yearly wages, and it have very high approval rate around 70%. Other wage units are rare. hourly wage positions face a high rejection rate of around 65%, while jobs with other wage units have a 36% rejection rate.

Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Preprocessing

Missing Data:

- Checked all there features and data seems normal and no missing values identified
- From statistical summary, it found that 33 employer have negative employee count.
- This negative number are mostly typographical error, and it fixed by replacing the negative values with absolute values
- Case ID is unique for each application and will be dropped
- No missing values were found across the dataset. This suggests a high level of data completeness and reliability for analysis.
- No duplicate values were found in the dataset.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Preprocessing

Categorical column are converted to 1 /0 using one hot encoding (**X = pd.get_dummies(X,drop_first=True)**) as there is no ordering involved in these feature variable values. Dropping first = True is applied to avoid multicollinearity as it is needed for regression models. After encoding data we get below feature sets for training

This resulted in 21 columns with most 18 columns as boolean and 3 numeric columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 21 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   no_of_employees                    25480 non-null  int64
 1   yr_of_estab                        25480 non-null  int64
 2   prevailing_wage                    25480 non-null  float64
 3   continent_Asia                     25480 non-null  bool
 4   continent_Europe                   25480 non-null  bool
 5   continent_North America            25480 non-null  bool
 6   continent_Oceania                  25480 non-null  bool
 7   continent_South America            25480 non-null  bool
 8   education_of_employee_Doctorate    25480 non-null  bool
 9   education_of_employee_High School  25480 non-null  bool
 10  education_of_employee_Master's     25480 non-null  bool
 11  has_job_experience_Y               25480 non-null  bool
 12  requires_job_training_Y            25480 non-null  bool
 13  region_of_employment_Midwest       25480 non-null  bool
 14  region_of_employment_Northeast     25480 non-null  bool
 15  region_of_employment_South         25480 non-null  bool
 16  region_of_employment_West          25480 non-null  bool
 17  unit_of_wage_Month                 25480 non-null  bool
 18  unit_of_wage_Week                  25480 non-null  bool
 19  unit_of_wage_Year                  25480 non-null  bool
 20  full_time_position_Y               25480 non-null  bool
dtypes: bool(18), float64(1), int64(2)
memory usage: 1.0 MB
```

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Preprocessing

Summary

- Case id alphanumeric unique identifier dropped
- Negative value of years of experience is fixed by using absolute values
- The dataset is clean and complete, with no duplicates or missing values.
- Outliers and unusual values were reviewed and deemed appropriate for the context, with no additional processing required at this stage.
- The data is suitable for further analysis or modeling without additional preprocessing.
- Categorical columns are hot encoded

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Preprocessing

When building the ML model it is essential to split the datasets in three parts: train, test and validation. This helps to ensure that model is trained verified and tested on independent data sets and prevents overfitting and provide realistic performance metrics

- Data is first split into 70% training and 30% test data sets
- Test data is further spit into 90% test and 10% validation

When splitting the data it is ensured that prediction data maintains its class ratio using `stratify=predicted_value`

Percentage of case_status classes and count

Case_status	training set	validation_set	test_Set
1	0.667919	0.66783	0.667974
0	0.332081	0.33217	0.332026

Case_status	training set	validation_set	test_Set
1	11913	5923	511
0	5923	2285	254

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- Checking the performance of the various tuned models. On Training and Validation dataset

Training performance comparison:

	Accuracy	Recall	Precision	F1
Gradient Boosting tuned with oversampled data	0.741753	0.891043	0.686167	0.775299
XGBoost tuned with oversampled data	0.814992	0.874759	0.781360	0.825426
AdaBoost tuned with oversampled data	0.650718	0.931923	0.596465	0.727380
Random forest tuned with undersampled data	0.500000	1.000000	0.500000	0.666667

Validation performance comparison:

	Accuracy	Recall	Precision	F1
Gradient Boosting tuned with oversampled data	0.722053	0.890292	0.743907	0.810543
XGBoost tuned with oversampled data	0.737171	0.852634	0.775951	0.812487
AdaBoost tuned with oversampled data	0.705480	0.930344	0.714716	0.808398
Random forest tuned with undersampled data	0.667830	1.000000	0.667830	0.800837

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- **Comparing the Performance of tuned Model**
- **Undersampled Data – Ensemble with Averaging/Voting (Random Forest)**

    Random Forest (undersampled) achieves the highest Recall (1.0) on both training and validation dataset, however it's Precision and Accuracy are the lowest among all models indicate poor balance

- **Oversampled Data – Ensemble with Boosting Algorithms**

Gradient Boosting: showed higher **Precision (0.77)** and the best **F1 Score (0.81)**, though its **Recall (0.85)** was lower than AdaBoost.

XGBoost: matched Gradient Boosting on **Precision (0.77)** and **F1 Score (0.81)**, while achieving slightly higher **Recall (0.89)** than Gradient Boosting, but lower than AdaBoost.

AdaBoost (Oversampled) delivered very high **Recall (0.93)** with improved **Precision (0.71)** and a balanced **F1 Score (0.80)**, outperforming Random Forest on overall stability.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

-

Final Model: AdaBoost(Oversampled Data – Ensemble with Boosting Algorithms)

As AdaBoost Model performed well on Recall (**0.93**) as well as on other parameter. So gives us very good overall performance that is generalized well as observed in train, test and validation dataset. It got much better precision score (0.71) vs Random forest which event though got perfect recall but precision score was much lower.

Models evaluation metrics on trian set

AdaBoostClassifier metrics train

	Accuracy	Recall	Precision	F1
0	0.650718	0.931923	0.596465	0.72738

Models evaluation metrics on Validation dataset

AdaBoostClassifier metrics validation

	Accuracy	Recall	Precision	F1
0	0.70548	0.930344	0.714716	0.808398

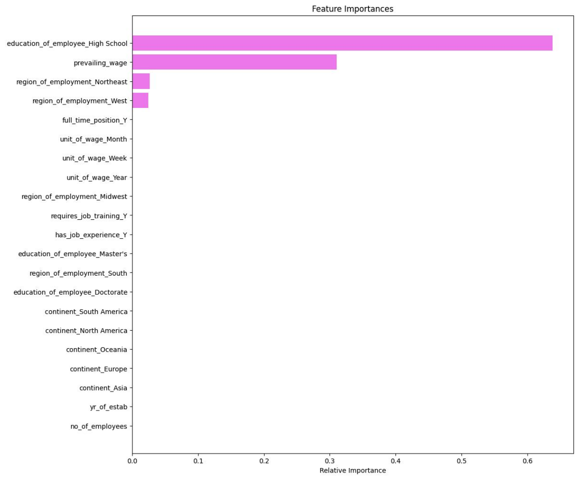
AdaBoost Models evaluation metrics on test set

	Accuracy	Recall	Precision	F1
0	0.71634	0.935421	0.722054	0.815004

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- Adaboost Model used the multiple features
    - Education : is very important factor in visa certification as candidate with higher education tend to have higer approval rate. And candidates with highschool only education have higher rejection rate
    - Wages: Prevailing wages are critical factor as very low wages have lower approval rate and certified candidates wages average is higher than who's visa got denied.
    - Region: Northeast and West region have large number of job opportunities and it seems these region got unique dynamics in job requirements



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

APPENDIX

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Data Overview

- Visa application Data had 25480 Rows with 12 columns

```
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
```

Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

- Sample rows from Top 5 and bottom 5 rows

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
0	EZYV01	Asia	High School	N	N	14513	2007	West	592.2029	Hour	Y	Denied
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast	83425.6500	Year	Y	Certified
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West	122996.8600	Year	Y	Denied
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West	83434.0300	Year	Y	Denied
4	EZYV05	Africa	Master's	Y	N	1082	2005	South	149907.3900	Year	Y	Certified

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
25475	EZYV25476	Asia	Bachelor's	Y	Y	2601	2008	South	77092.57	Year	Y	Certified
25476	EZYV25477	Asia	High School	Y	N	3274	2006	Northeast	279174.79	Year	Y	Certified
25477	EZYV25478	Asia	Master's	Y	N	1121	1910	South	146298.85	Year	N	Certified
25478	EZYV25479	Asia	Master's	Y	Y	1918	1887	West	86154.77	Year	Y	Certified
25479	EZYV25480	Asia	Bachelor's	Y	N	3195	1960	Midwest	70876.91	Year	Y	Certified

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Statistical summary of the data

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.0	NaN	NaN	NaN	5667.04321	22877.928848	-26.0	1022.0	2109.0	3504.0	602069.0
yr_of_estab	25480.0	NaN	NaN	NaN	1979.409929	42.366929	1800.0	1976.0	1997.0	2005.0	2016.0
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.0	NaN	NaN	NaN	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results

Statistical Insight

- Visa application Data had 25480 Rows with 12 columns with 3 numeric and 9 categorical or alphanumeric columns
- Number of employees have wide range of spectrums as mean is around 5000 and max employee count is 602069. Number of employee column have some negative value and indicates there are some data anomaly that needs to be addressed
- Prevailing wages are had wide range where min is 2 to max being 319210.27. This indicates prevailing wages might not be captured correctly or have to computed based on some other factors
- Yearly unit is used in most of the job around 90%, but there are some data samples with other units too and may need some standardization
- Some employers are well established and more than century old and some are start up. But 75% organization are established before 2005 ands mean is around 1979.
- Case number is alphanumeric and unique for each row and will be dropped

Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

EDA Results

continent: There are six continents and 66% candidates are from Asia contribute and very few applicant are from South America and Africa

education_of_employee: Doctorate candidates are very less, also candidate with High-school are also small part of the applicant pool. Candidates with Bachelor's and Master’s degree are major part of the candidates pool

has_job_experience:60% of the applicants have the prior job experience and 40 percent of the candidates have no prior job experience. It indicates job applicant are well represented from various experience background

requires_job_training: 88% of the job don’t need on the job training and only 11% job need the job training

region_of_employment: Island represent very small (375/25480) job position and seems ok as it is inline with their geographical size. Midwest also represents only 17% of the job location, other region have very similar distribution of the opportunities

unit_of_wage:90% of the job listed with Yearly wages and very few jobs listed with other unit of wages. Need to check with business if unit of wage can be converted to same unit for ease of comparison and analysis

full_time_position: Around 90% of job offers full-time positions and only 10% of the job are not offering full-time position. This seems normal but need to consult with business for if any further analysis is needed.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Preprocessing

Prevailing Wages:

- There are numerous entries with prevailing wages above $200,000.
- These high values are considered legitimate outliers, reflecting real scenarios where certain positions command very high salaries.

Number of Employees

- The range of employee counts is wide, which is expected due to the diversity of employers (from small businesses to large corporations). From statistical summary, it found that 33 employer have negative employee count
- Updated -ve values with absolute values and

Year of Establishment

- The years of establishment span a large range, consistent with a mix of both new and long-standing organizations.
- The data appears normal, so no special treatment done.

Case_status

- Replaced Y and N with 1 and 0 as it is prediction value and dropped from the training set

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Statistical Insight - Categorical column's insight by checking unique categorical data .

Case number is alphanumeric and unique for each row and will be dropped

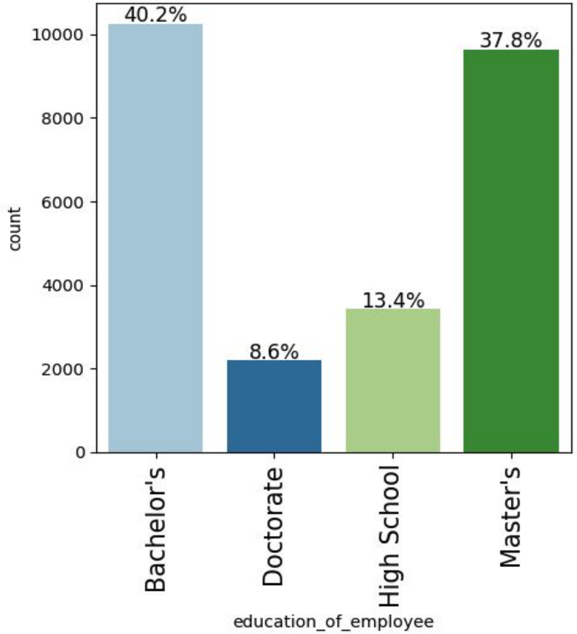
```
--------------------------------------------------
continent
Asia             16861
Europe            3732
North America     3292
South America      852
Africa             551
Oceania            192
Name: count, dtype: int64
--------------------------------------------------
education_of_employee
Bachelor's     10234
Master's        9634
High School     3420
Doctorate       2192
Name: count, dtype: int64
--------------------------------------------------
has_job_experience
Y    14802
N    10678
Name: count, dtype: int64
--------------------------------------------------
requires_job_training
N    22525
Y     2955
Name: count, dtype: int64
--------------------------------------------------
region_of_employment
Northeast    7195
South        7017
West         6586
Midwest      4307
Island        375
Name: count, dtype: int64
--------------------------------------------------
unit_of_wage
Year     22962
Hour      2157
Week       272
Month       89
Name: count, dtype: int64
--------------------------------------------------
full_time_position
Y    22773
N     2707
Name: count, dtype: int64
--------------------------------------------------
case_status
Certified    17018
Denied        8462
Name: count, dtype: int64
--------------------------------------------------
```

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Data Overview

- **education_of_employee**: applicant with bachelor's and master's degree are most common and Doctorate or High school applicant are very rare.



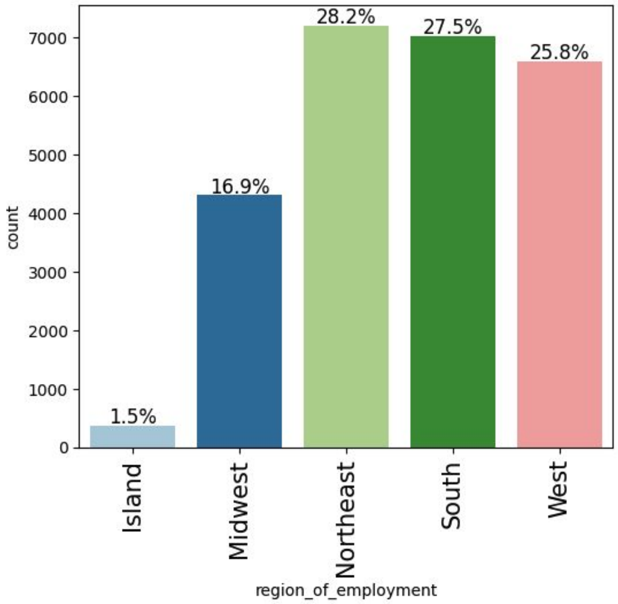
Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Data Overview

- **region_of_employment**: The Northwest, South, and West regions collectively account for 81% of job opportunities with similar distribution patterns, while the Midwest offers fewer positions and the Island region has only a rare opportunities

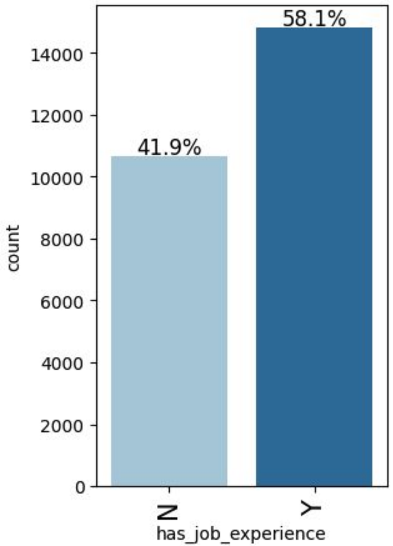


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Data Overview

- **has_job_experience**: applicants with prior job experience make up a larger share of the application pool at around 58%, while 41% of candidates have no previous experience.

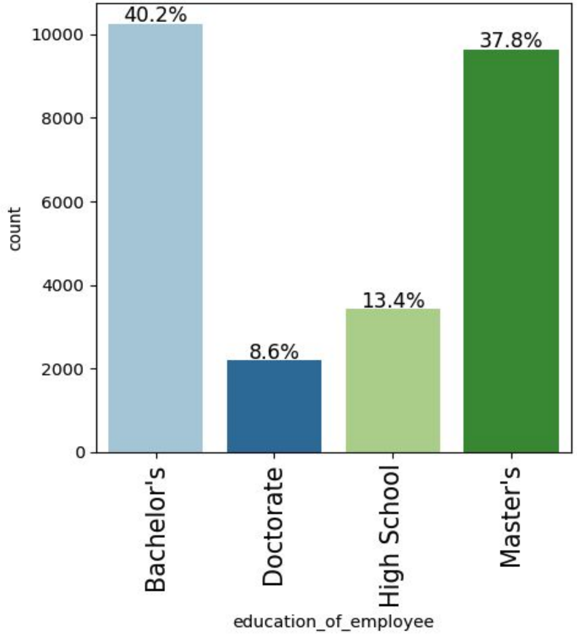


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Data Overview

- **education_of_employee**: applicant with bachelor's and master's degree are most common and Doctorate or High school applicant are very rare.

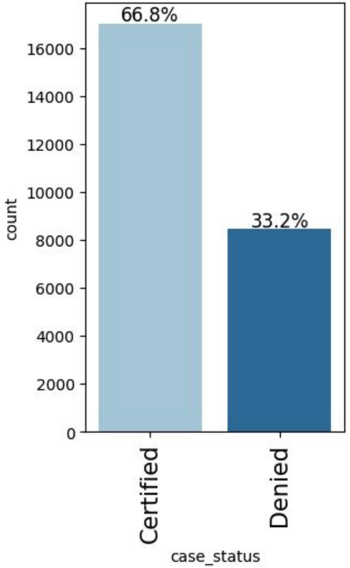


Link to Appendix slide on data background check

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

- **case_status**: Approximately 66% of applicants are approved, while one-third face rejection; the machine learning model aims to predict the case status based on key applicant features



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

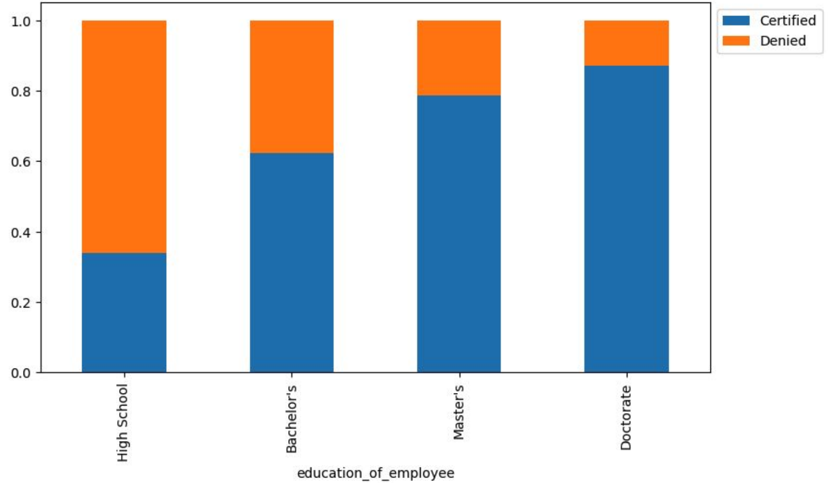
Correlation metrics reveals negligible correlation among prevailing wage, year of establishment and number of employees as correlation coefficient is near zero. These Features are statistical independent and each provide unique information and can't be used as substitute for others in the model building

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Education of employee and case status: Approximately two-thirds of applicants with education up to high school had their visas denied. In contrast, candidates with a Doctorate represent a small portion of the applicant pool and face only a 12% rejection rate. Master’s degree holders experience relatively low rejection, while Bachelor’s degree applicants face a moderate denial rate compared to higher education levels.

case_status	Certified	Denied	All
education_of_employee			
All	17018	8462	25480
Bachelor's	6367	3867	10234
High School	1164	2256	3420
Master's	7575	2059	9634
Doctorate	1912	280	2192

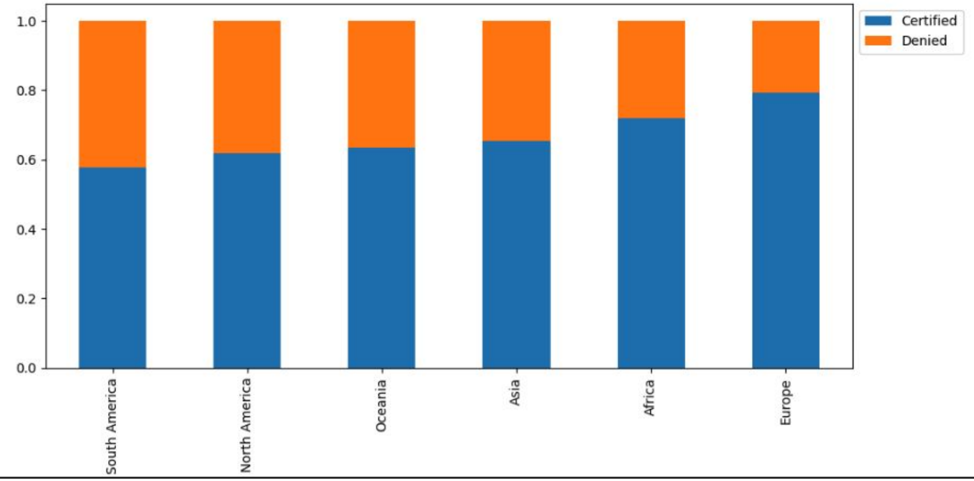


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Continent of employee and case status: Region plays a significant role in visa approval outcomes. Applicant from Europe have the highest approval rate, around 80%, followed by those from Africa and Asia. In contrast, applicants from South America face the lowest certification rates. Additionally, candidates from Oceania and Africa are relatively rare in the applicant pool.

case_status continent	Certified	Denied	All
All	17018	8462	25480
Asia	11012	5849	16861
North America	2037	1255	3292
Europe	2957	775	3732
South America	493	359	852
Africa	397	154	551
Oceania	122	70	192

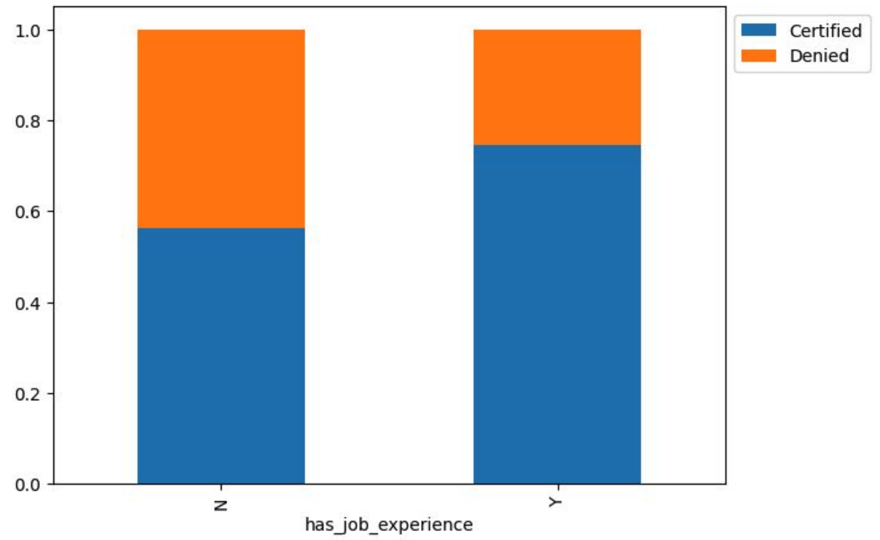


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Prior Job experience: Applicants with prior job experience not only form the majority of the applicant pool but also have a significantly higher approval rate of around 75%, compared to just 56% for those without prior experience

case_status has_job_experience	Certified	Denied	All
All	17018	8462	25480
N	5994	4684	10678
Y	11024	3778	14802

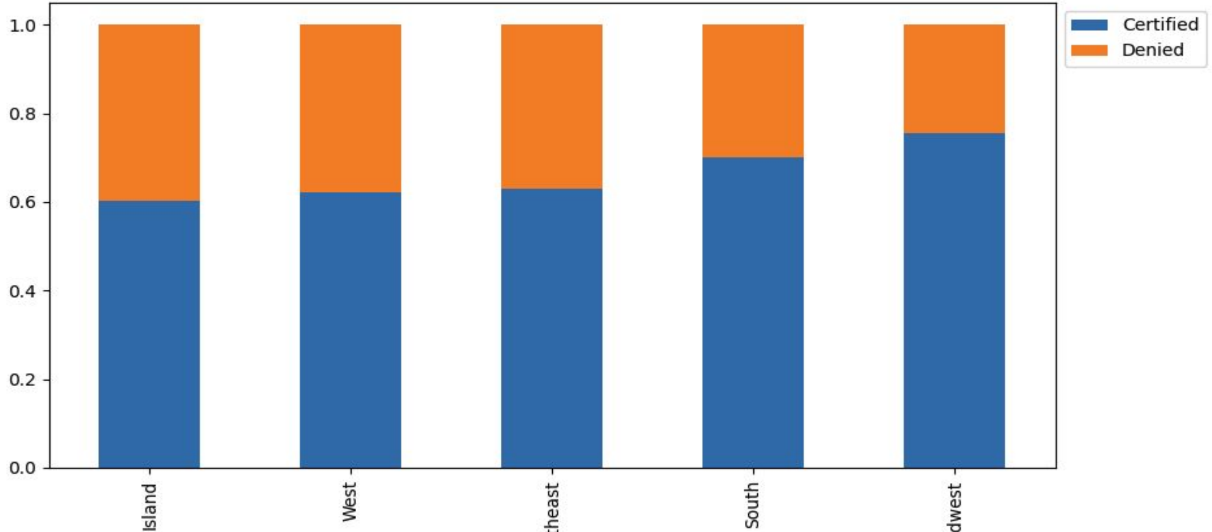


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

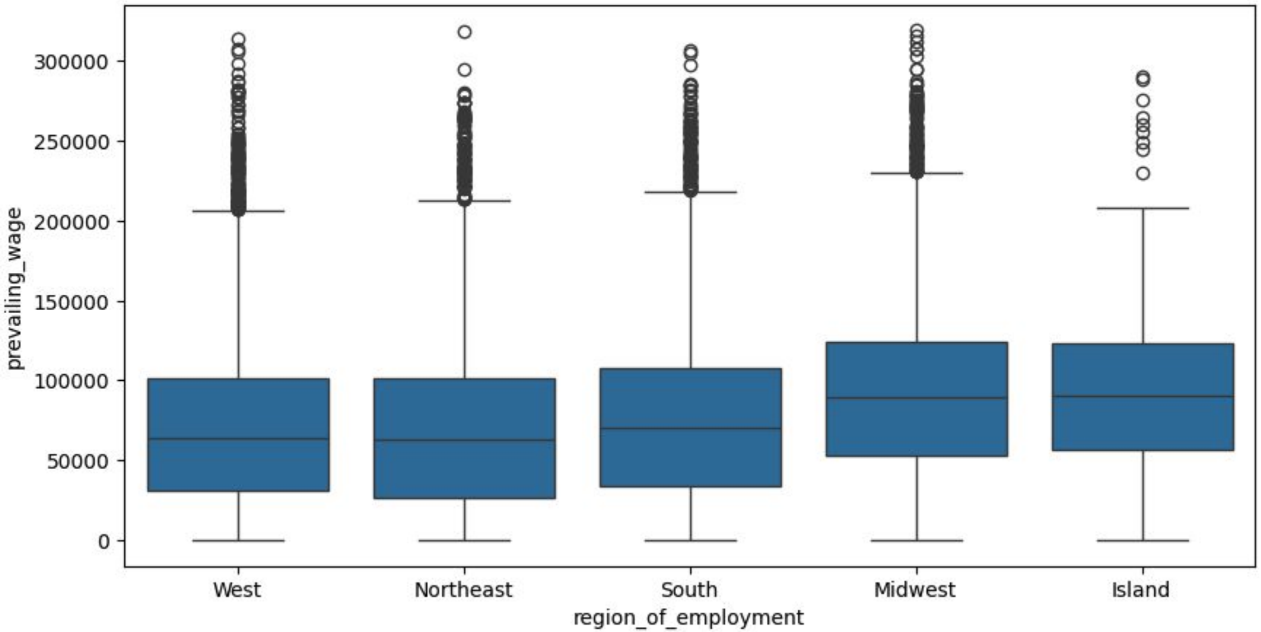
Prior Job experience: Employment Region have little impact on the case status but Midwest has lowest rejection rate and Island have higher rejection rate but data small data could be contributing factor in the skewed results

case_status region_of_employment	Certified	Denied	All
All	17018	8462	25480
Northeast	4526	2669	7195
West	4100	2486	6586
South	4913	2104	7017
Midwest	3253	1054	4307
Island	226	149	375



Data Background and Contents

Prevailing wages vs region of employment: Across all regions of employment, prevailing wages show significant outliers that impact the mean. The Island and Midwest regions have the highest average wages (~91K), while the Northwest has the lowest (~67K). Interestingly, the South reports higher average wages than the West. Overall, wages range widely—from below 100 to over 300K

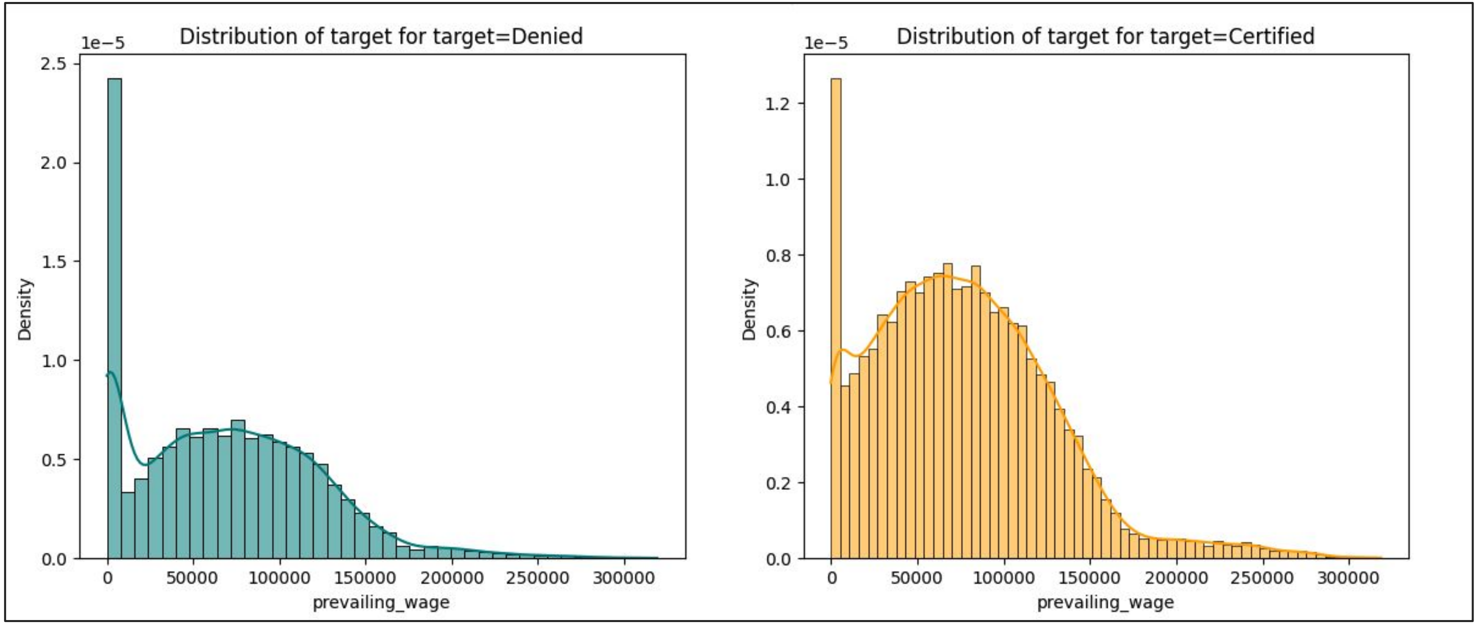


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Prevailing wages and approval status: Certified candidates show a strong concentration of wages between 60K and 120K, with a right-skewed distribution and a mean of 77K (excluding outliers). Lower-wage applicants are rarely certified, and a noticeable spike at $0 may indicate internships or data anomalies.

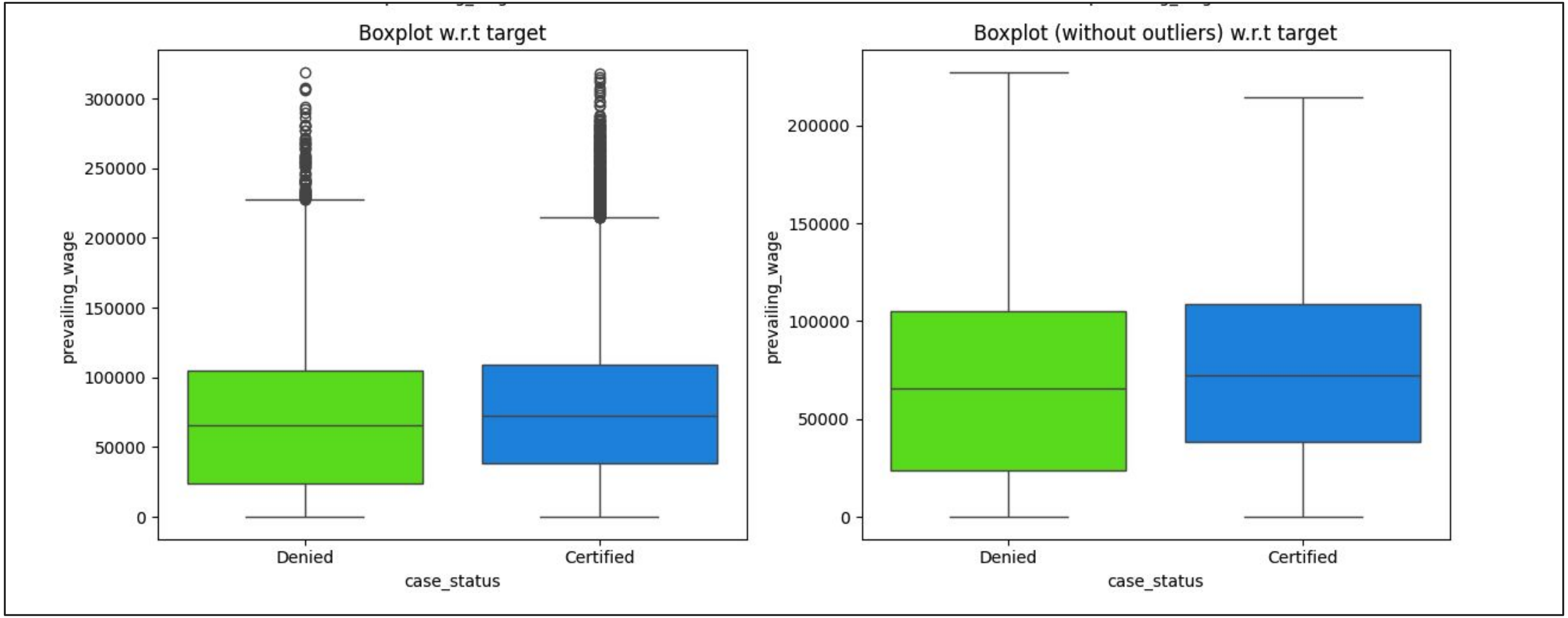
Denied candidates tend to have lower wages, with a flatter, slightly right-skewed distribution and a mean of 68K. A significant spike near $0 is also observed in denied cases, and wages above 200K are rare among them.



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Case_status: Certified applicants tend to have higher wages, a trend clearly reflected in the box plot—with or without outliers. The mean wage for certified candidates is approximately 77K, compared to 68K for denied applicants. When outliers are removed, the mean drops slightly to around 73K for certified and 66K for denied applicants, reinforcing the positive correlation between higher wages and visa approval.

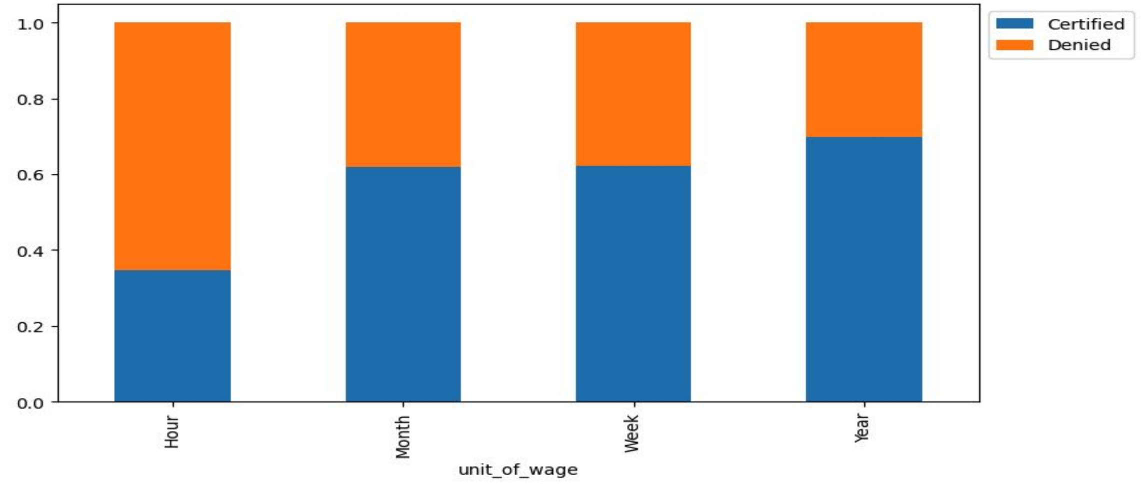


Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Unit of wages and case status: Approximately 90% of jobs offer yearly wages, and it have very high approval rate around 70%. Other wage units are rare. hourly wage positions face a high rejection rate of around 65%, while jobs with other wage units have a 36% rejection rate.

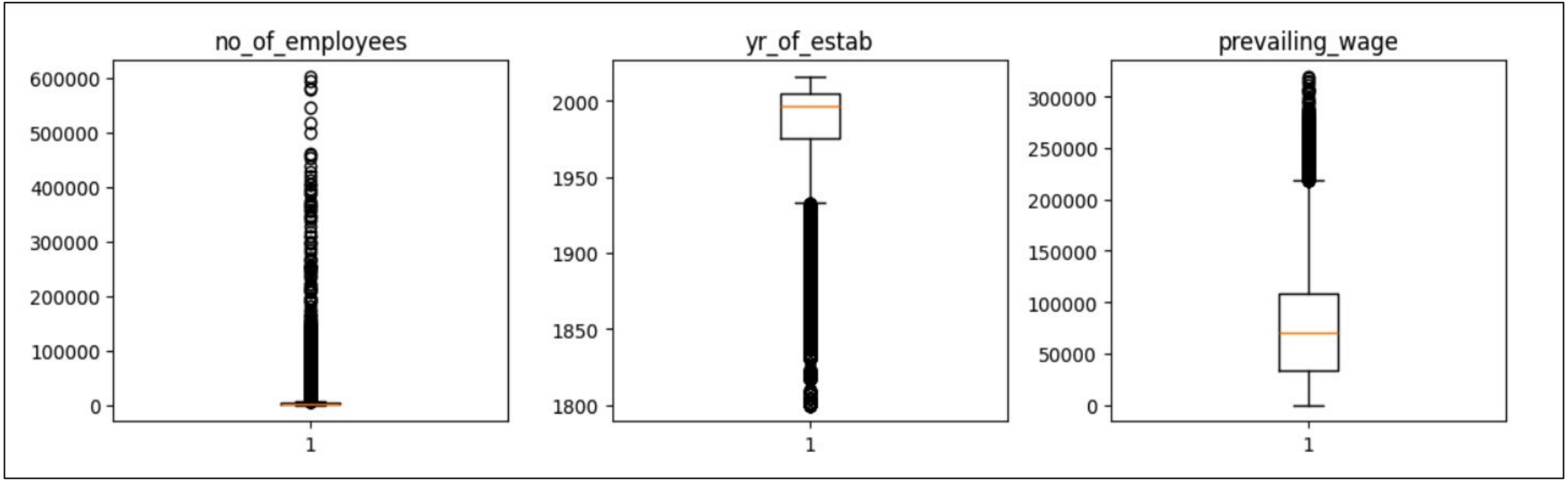
```
case_status    Certified  Denied    All
unit_of_wage
All                17018    8462  25480
Year               16047    6915  22962
Hour                 747    1410   2157
Week                 169     103    272
Month                 55      34     89
------------------------------------------------------------------------------------------------------------------------
```



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

- Employee counts vary widely, from as few as 11 to over 600,000, though 75% of employers have fewer than 3,500 employees, with notable outliers.
- Most organizations were established before 2000, with an average founding year around 1979, and some dating back over a century.
- Prevailing wages typically fall below $100K, but there are significant outliers reaching as high as $319K.



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Data Background and Contents

Case_status: Replaced Y and N with 1 and 0 as it is prediction value and dropped from the training set

Categorical column are converted to 1 /0 using one hot encoding (**X = pd.get_dummies(X,drop_first=True)**)as there is no ordering involved in these feature variable values. Dropping first = True is applied to avoid multicollinearity as it is needed for regression models. After encoding data we get below feature sets for training

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 21 columns):
 #   Column                             Non-Null Count  Dtype  
---  ------                             --------------  -----  
 0   no_of_employees                    25480 non-null  int64  
 1   yr_of_estab                        25480 non-null  int64  
 2   prevailing_wage                    25480 non-null  float64
 3   continent_Asia                     25480 non-null  bool   
 4   continent_Europe                   25480 non-null  bool   
 5   continent_North America            25480 non-null  bool   
 6   continent_Oceania                  25480 non-null  bool   
 7   continent_South America            25480 non-null  bool   
 8   education_of_employee_Doctorate    25480 non-null  bool   
 9   education_of_employee_High School  25480 non-null  bool   
 10  education_of_employee_Master's     25480 non-null  bool   
 11  has_job_experience_Y               25480 non-null  bool   
 12  requires_job_training_Y            25480 non-null  bool   
 13  region_of_employment_Midwest       25480 non-null  bool   
 14  region_of_employment_Northeast     25480 non-null  bool   
 15  region_of_employment_South         25480 non-null  bool   
 16  region_of_employment_West          25480 non-null  bool   
 17  unit_of_wage_Month                 25480 non-null  bool   
 18  unit_of_wage_Week                  25480 non-null  bool   
 19  unit_of_wage_Year                  25480 non-null  bool   
 20  full_time_position_Y               25480 non-null  bool   
dtypes: bool(18), float64(1), int64(2)
memory usage: 1.0 MB
```

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Original Data

In this solution following Classifier algorithm are used with Random state =1

- **DecisionTree**
- **Ensemble Models :** (Voting for Classification, Averaging for Regression)
    - Bagging:
        - Trains multiple base models on different subsets of the training data
    - RandomForest
        - Trains multiple base model (weak trees) on different subsets of features.
- **Boosting based Ensemble Models:** Train sequentially to reduce error made in previous stages
    - Gradient Boost**:**
        - Builds models sequentially using gradient descent to minimize prediction errors at each stage
    - AdaBoost:
        - Focus on previously misclassified instances to improve accuracy.
    - XGB (extreme gradient boost):
        - optimized Gradient boost model to improve speed and accuracy

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Original Data

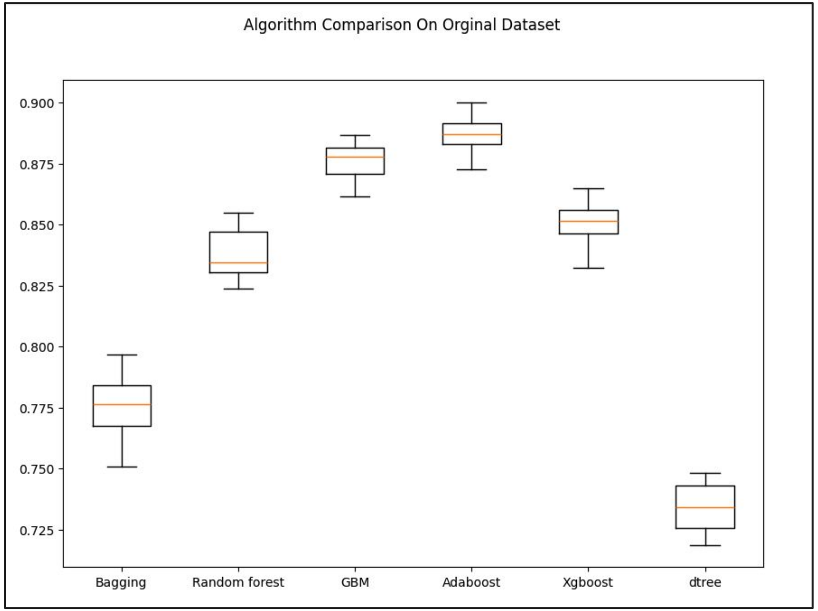
- All the models are train with 10 fold stratified cross-validation to evaluate model performance using Recall score. StratifiedKFold is used to ensure that each fold in cross-validation preserves the original class distribution
- Models achieved below cv scores on Training dataset.

Model	Cross validation score
Bagging	0.7762967784107879
Random forest	0.8376582759961455
GBM	0.8760198834660402
AdaBoost	0.8873517263142471
XgBoost	0.851089336128345
DecisionTree	0.7340732929859856

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Original Data

- **Comparing performance**
    - Adaboost Model performed the best followed by GBM , and XGBoost
    - Bagging and DecisionTree didn’t performed well and Random was better comparing these two



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

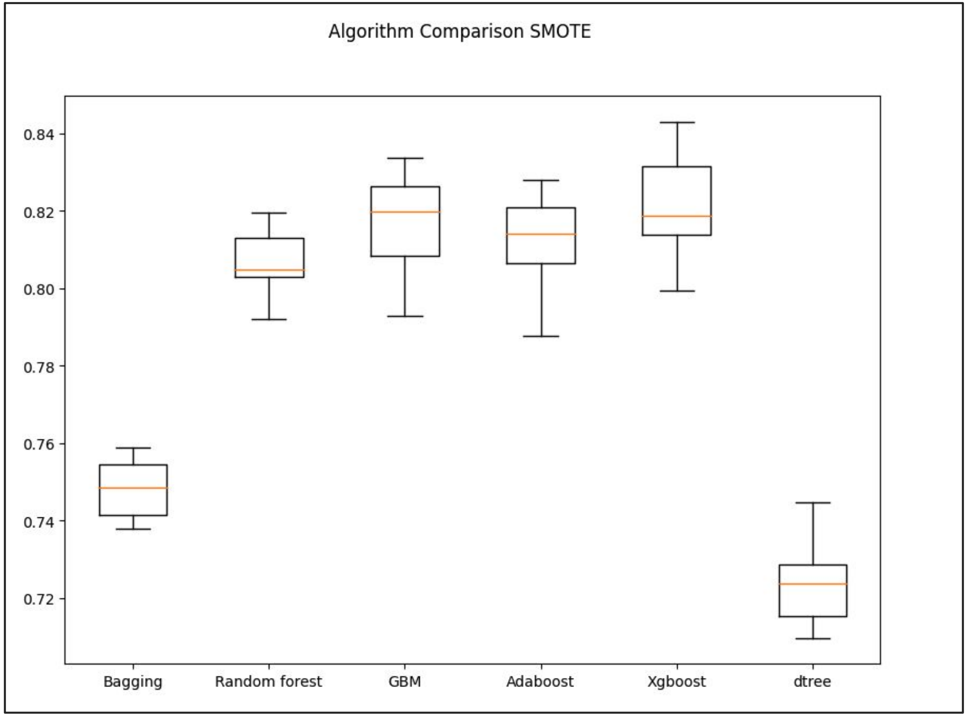
Model Building - Oversampled Data

- **Oversampling** applied to balance the dataset by increasing the representation of the minority class.
- For Oversampling SMOTE (Synthetic Minority Oversampling Technique) is used. IT generates synthetic samples for the minority class to balance the dataset. It works by interpolating between existing minority class instances and their nearest neighbors.This helps improve model performance on imbalanced datasets by reducing bias toward the majority class.
- Oversampling increased dataset volume approximately 33% and
- class 0 representation increased from **33 % of total volume** to **50% of total** volume so both class gets similar weights
  - Before OverSampling, counts of label '1': 11913
  - Before OverSampling, counts of label '0': 5923
  - After OverSampling, counts of label '1': 11913
  - After OverSampling, counts of label '0': 11913
  - After OverSampling, the shape of train_X: (23826, 21)
  - After OverSampling, the shape of train_y: (23826,)

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Oversampled Data

- All the models are trained on the oversampled dataset and collected cross validation score as per the below boxplots.



Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Oversampled Data

- All the models are trained on the oversampled dataset and collected cross validation score as per the below boxplots.
  Got below Validation Performance score achieved on oversampled data.

Model	SMOTE	Original
Bagging:	0.7483437019255151	0.7762967784107879
Random forest:	0.8062643342969361	0.8376582759961455
GBM:	**0.8175980085540886**	0.8760198834660402
Adaboost:	0.812055953769603	**0.8862601361441234**
Xgboost:	0.8213748668706573	0.851089336128345
dtree:	0.7539503386004515	0.7340732929859856

- Comparison of box plots and cv scores shows:
  - **Gradient Boost** performed the best among all models.
  - **AdaBoost** and **XGBoos**t performance slightly declined on SMOTE data, possibly due to overfitting or loss of natural distribution
  -
  - **Random Forest ,DecisionTree and Bagging models** performance declined slightly on the oversampled data.
- **Overall**, most models performed similarly or worse on the oversampled data, suggesting a need for **hyperparameter tuning** to fully leverage SMOTE.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Oversampled Data

- Computing Accuracy, Precision , Recall and F1 score models for Oversampled data sets.
  - **GBM and XGBoost** achieved very close recall scores, but **GBM slightly outperformed XGBoost** with better precision and F1 score
    AdaBoost performed well overall, with above-average metrics across the board
  - **RandomForest** showed good recall but slightly lower precision and accuracy than boosting models.
  - **Bagging** had moderate performance across all metrics.
  - **DecisionTree** recorded the lowest recall, precision, and F1 scores, indicating it is the weakest model in this setup.

SMOTE metrics for evaluaiton

	Accuracy	Recall	Precision	F1
Bagging	0.687891	0.746408	0.777375	0.761577
Random forest	0.711586	0.799303	0.775665	0.787307
GBM	0.729903	0.815847	0.787395	0.801368
Adaboost	0.704027	0.807357	0.763169	0.784641
Xgboost	0.716529	0.814758	0.773028	0.793345
dtree	0.659253	0.727035	0.753950	0.740248

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Undersampled Data

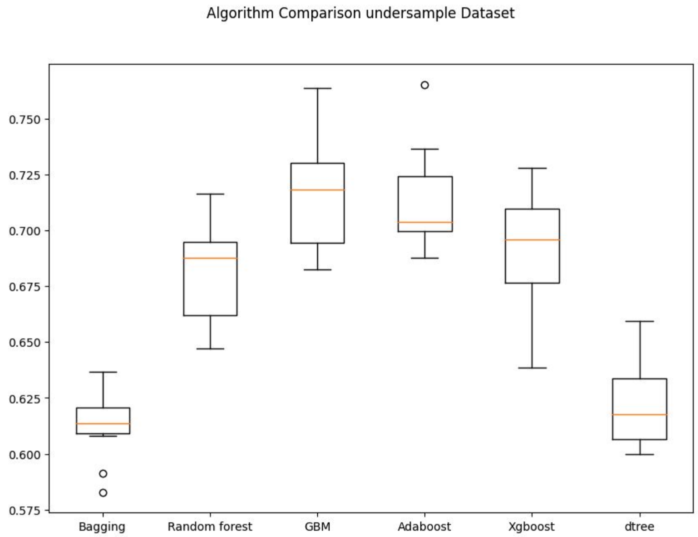
- **Undersampling** applied to balance the dataset by reducing the size of the majority class, ensuring equal representation of both classes.
  - This solution uses **RandomUnderSampling** to reduce the majority class size.
  - Undersampling helps **reduce training time** by decreasing data volume.
  - However, it may cause **loss of information**, so it must be used cautiously

- Checking the data before and after undersampling and observed that class imbalance is addressed correctly and improved class 0 representation from 33% to 50%
- It also reduced data volume by 33%
  - Before UnderSampling, counts of label '1': 11913
  - Before UnderSampling, counts of label '0': 5923
  - After UnderSampling, counts of label '1': 5923
  - After UnderSampling, counts of label '0': 5923
  - After UnderSampling, the shape of train_X: (11846, 21)
  - After UnderSampling, the shape of train_y: (11846,)

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Undersampled Data

- All the models are trained on the undersampled dataset and collected cross validation score and build boxplots on score.
- Comparing scores GBM and Adaboost both performed similar on Recall but GBM has better F1 and Precision score
- XGBoost and Random Forest performed very close to each other for Recall and RandomForest got better precision and XGboost got better F1 score
- DecisionTree and Bagging showed weakest performance on all metrics on undersampled data



Cross Validation Score
Bagging: 0.6133710860945263
Random forest: 0.6819168451756984
GBM: 0.7182164668884736
Adaboost: 0.7134901554167995
Xgboost: 0.6922055170685019
dtree: 0.6212974568160065

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Building - Undersampled Data

- **GBM and AdaBoost** had similar **Recall**, but **GBM** outperformed AdaBoost with higher **Precision** and **F1 Score**.
- **XGBoost and Random Forest** had comparable **Recall**, with **Random Forest** achieving better **Precision**, while **XGBoost** had a higher **F1 Score**.
- **Random Forest** stood out with better **Precision** than XGBoost and comparable **Recall** and **F1 Score**.
- DecisionTree and Bagging showed weakest performance on all metrics on undersampled data

under sampled data evaluation

	Accuracy	Recall	Precision	F1
Bagging	0.639482	0.605572	0.806377	0.691696
Random forest	0.675825	0.670222	0.811545	0.734144
GBM	0.702573	0.708533	0.821555	0.760870
Adaboost	0.694432	0.708533	0.810105	0.755922
Xgboost	0.682367	0.685242	0.809879	0.742365
dtree	0.616223	0.621680	0.759979	0.683908

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Improvement - Hyperparameter Tuning

- Gradient Boost, AdaBoost, and XGBoost performed well on oversampled data, configuration.
- Random Forest showed strong, consistent performance on undersampled data and performed comparably to XGBoost, making it a strong candidate among non-boosting ensemble models.
- DecisionTree , and Bagging didn't faired well in any of the training and validation metrics .

Got below Validation Performance score achieved on oversampled data.

Model	SMOTE	Original	Undersample
Bagging:	0.7483437019255151	0.7762967784107879	0.6133710860945263
Random forest:	0.8062643342969361	0.8376582759961455	**0.6819168451756984**
GBM:	**0.8175980085540886**	0.8760198834660402	0.7182164668884736
Adaboost:	**0.812055953769603**	0.8862601361441234	0.7134901554167995
Xgboost:	**0.8213748668706573**	0.851089336128345	0.6922055170685019
dtree:	0.7539503386004515	0.7340732929859856	0.6212974568160065

- Considering all the above points below models are tuned further
  a. GradinetBoost with OverSampled data
  b. Adaboost with Oversampled data
  c. XGBoost with Oversampled Data
  d. RandomForest on Undersampled Data

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Improvement - Hyperparameter Tuning

Model Tuning

- DecisionTree and Bagging showed consistently weak performance on both oversampled and undersampled data. These models will be excluded from further tuning. Focus will shift to models that performed well or had comparable results
- Hyperparameters can be tuned with help of RandomizedSearchCV or GridSearchCV
- **GridSearchCV** tries all the possible combination of the given parameter grid and very expensive and in some cases not helpful
- **RandomizedSearchCV** is faster and tries n random sample and can scale better. Considering this flexibility models are tuned with RandomizedSearchCV

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Improvement - Hyperparameter Tuning

- **AdaBoost with OverSampled Data**
- Following ParameterGrid Supplied to Tuned the AdaBoost with OverSampled Data

```
param_grid = {
  "n_estimators": [50,75,100,125],
  "learning_rate": [1.0, 0.5, 0.1, 0.01],
        "estimator": [DecisionTreeClassifier(max_depth=1, random_state=1), DecisionTreeClassifier(max_depth=2,
random_state=1), DecisionTreeClassifier(max_depth=3, random_state=1),]}
```

N_estimator: Indicates number of weak learners in the ensemble higher value get better result but time consuming so 50-125 range seems good range for tuning

learning_rate: Is a scaling factor which impact the contribution of each weak learner low value is slow but generalize better and higher value learn faster but risk of overfitting

Estimator: Indicates the algorithm to use DecisionTree is used with Tree depth value range from 1 to 3

Overall 48 combinations (4 estimators , 4 learning rate and three depth) will be used to find best parameters

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Improvement - Hyperparameter Tuning

Tuned AdaBoost : cv score 0.931923507999745

```
AdaBoostClassifier
AdaBoostClassifier(estimator=DecisionTreeClassifier(max_depth=1,
                                                    random_state=1),
                   learning_rate=0.1, random_state=1)
estimator: DecisionTreeClassifier
DecisionTreeClassifier(max_depth=1, random_state=1)
DecisionTreeClassifier
DecisionTreeClassifier(max_depth=1, random_state=1)
```

- The tuned model shows consistent performance on both the training and validation datasets, indicating good generalization with no signs of overfitting and it achieve very good Recall score on Training and validation score. Precision score improved in validation dataset indicate good generalization of model.

AdaBoostClassifier metrics train

	Accuracy	Recall	Precision	F1
0	0.650718	0.931923	0.596465	0.72738

AdaBoostClassifier metrics validation

	Accuracy	Recall	Precision	F1
0	0.70548	0.930344	0.714716	0.808398

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- **RandomForest**:Tuned using undersampled data with the help of RandomizedSearchCV, leveraging the following hyperparameters:
  param_grid = {
  "n_estimators": [50, 75, 100, 125],.
  "min_samples_leaf": [1, 2, 4, 5, 10],
  "max_features": ['auto', 'sqrt', 'log2', None],
  "max_samples": [0.5, 0.6, 0.7, 0.8, 0.9, 1, None],)

- **n_estimators**: number of estimators in ensembles or number of decisiontree in the RandomForest
- **min_samples_leaf**: minimum number of samples in the leaf node higher values avoid overfitting
- **max_features**: Number of features to consider when looking for the best split; sqrt is square root of total feature e.g. 64 features will use 8 features at each split. Or log2 uses log of the feature $\log_2(100)$ ~ 6~7 features at split
- **max_samples:** Fraction of the training sets to use for each tree allows to train tree on subset of the data

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

Tuned RandomForest cv score=1.0

```
RandomForestClassifier
RandomForestClassifier(max_features='log2', max_samples=1, min_samples_leaf=2,
                       random_state=1)
```

Tuned Random Forest showed perfect recall score on the training and validation dataset, but lowest precision score. Model generalized well on the validation dataset. And shown improvement compared to training data set on all aspects

RandomForestClassifier metrics trained data

	Accuracy	Recall	Precision	F1
0	0.5	1.0	0.5	0.666667

RandomForestClassifier metrics validaiton d

	Accuracy	Recall	Precision	F1
0	0.66783	1.0	0.66783	0.800837

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

GradientBoosting Tuned with Oversampled Data with RandomizedSearchCV with below hyperparameter range

```
param_grid={
  "n_estimators": [100, 125, 150, 175], ## Complete the code to set the number of estimators.
  "learning_rate": [0.1, 0.05, 0.01, 0.005], ## Complete the code to set the learning rate.
  "subsample":[0.7, 0.8, 0.9, 1.0], ## Complete the code to set the value for subsample.
  "max_features":['auto', 'sqrt', 'log2', 0.3, 0.5, None] ## Complete the code to set the value for max_features.
}
```

- **n_estimators**: number of tree to build sequentially and higher estimator improve performance but time consuming and may overfit
- **learning_rate**: controls the contribution of each Tree to the final Model lower value train slow and generalize well. Higher value will speed up the training but may overfit the model
- **subsample** Fraction of training dataset used for each tree to reduce overfitting
- **max_feature:** decides on number of feature to use for each split; sqrt and log2 function applied on total feature to get value of features used for the each split

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

Tuned **GradientBoosting** : cv score=0.8917154924346924

```
GradientBoostingClassifier
GradientBoostingClassifier(learning_rate=0.01, max_features=0.5,
                           n_estimators=125, random_state=1, subsample=0.7)
```

Model performed very similar on Train and Validation dataset. Precision is improved in the Validation st but recall is deteriorated slightly but over all model generalized well

GradientBoostingClassifier metrics trained data

	Accuracy	Recall	Precision	F1
0	0.741753	0.891043	0.686167	0.775299

GradientBoostingClassifier metrics validation data

	Accuracy	Recall	Precision	F1
0	0.722053	0.890292	0.743907	0.810543

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- XGBoost Tuned with OVersampled Data

```
param_grid={'n_estimators':[50,75,100,125],'subsample':[0.5, 0.7, 0.8, 1.0],
            'gamma':[0,1, 3, 5,8], 'colsample_bytree':[0.3, 0.5, 0.7, 1.0],
            'colsample_bylevel':[0.3, 0.5, 0.7, 1.0], }
```

n_estimator: Number of boosting rounds(tree) to be tried higher value give better learning but there is risk of Overfitting

Subsample: fraction of the rows to train each tree small fraction add random ness and 1 values means all data is used for training each tree.

Gamma: minimum loss reduction at each split it control number of splits

colsample_bytree: fraction of features to use for each tree to build

colsample_bylevel: fraction of features to use at each tree level not just per tree works similar to colsample_bytree but at much granular level.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- Tuned **XGBoost** :cv score=0.8917154924346924

XGBClassifier

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=0.5, colsample_bynode=None,
              colsample_bytree=1.0, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, gamma=1, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=100,
              n_jobs=None, num_parallel_tree=None, random_state=1, ...)
```

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Model Performance Summary and Final Model Selection

- Tuned XGBoost performance result on validation datasets have similar or better performance compare to Training data set
- Recall score remain same and precision and F1 score improved on validation dataset. But accuracy score was deteriorated. Overall model performed well and generalized

GradientBoostingClassifier metrics trained data

	Accuracy	Recall	Precision	F1
0	0.741753	0.891043	0.686167	0.775299

GradientBoostingClassifier metrics validation data

	Accuracy	Recall	Precision	F1
0	0.722053	0.890292	0.743907	0.810543

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Key Takeaway

- Startups and small organizations face lower approval rates; targeted policy support or improved evidence submission may help level the field
- Geographic origin strongly impacts visa outcomes (e.g., 42% denial for South America vs. 20% for Europe); further investigation is warranted.
- Candidates aiming for approval should focus on high-wage positions and well-established companies, as both factors correlate strongly with certification.
- Applicants with higher wages and prior job experience have a better chance of certification
- Standardizing wage units (e.g., converting all to yearly wages) can improve fairness and reduce comparison bias.
- Full-time, high-paying positions offered by well-established employers are more likely to get certified.

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Key Takeaways

01 —— Education and wages highly influence visa certification.

02 —— Full-time, yearly wage jobs are most often certified.

03 —— Europe has the highest visa approval rates by continent.

04 —— Prior job experience increases visa approval chances.

This is AI generated summary Attached just for ref:

Proprietary content. © Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited.

Great Learning

Happy Learning !